

YallaJo

Complete Business Rules & Edge Cases

Version 1.0

Phase 1: Core Features & Foundation

Business Rules & Edge Cases

1. Service Provider Registration

1.1 Provider Types

Provider Type	Description	Required Documents
Independent Tour Guide	Individual licensed guide offering tours	Government ID, MoTA License, Insurance, TIN
Business Owner	Hotel, restaurant, cafe, or shop owner	Business License, Tax Certificate, Proof of Ownership/Lease
Agency	Company managing multiple tour guides	Business License, Tax Certificate, Proof of Ownership/Lease + list of affiliated guides
Freelance Activity Instructor	Yoga, diving, cooking class, etc.	Government ID, Insurance, TIN + relevant certification

1.2 Registration Workflow

Register/Login → Verify OTP → Select Provider Type → Upload Required Documents → Status: "Pending" → Admin Review (within 7-day SLA) → Approve / Reject (with reason) / Request More Documents → If approved: dashboard access granted, provider can create listings.

1.3 Business Rules

Registration Rules

- ✓ A user account must be verified (OTP) before starting the provider application.
- ✓ All required documents for the selected provider type must be uploaded before submission. Partial submissions are saved as draft but cannot be submitted for review.
- ✓ Maximum file size: 5 MB per document.
- ✓ Accepted formats: PDF, JPG, PNG.
- ✓ Maximum 10 documents per application (including re-submissions).
- ✓ A single user can hold only ONE provider type at a time. To switch types (e.g., from Tour Guide to Agency), the previous provider profile must be deactivated first.
- ✓ Agencies must list at least one affiliated tour guide during registration.

Admin Review Rules

- ✓ Admin has a 7-day SLA to review new applications. System sends an internal reminder to admin at day 5 if still pending.

- ✓ Admin can: Approve, Reject (must provide reason), or Request More Documents (provider gets notified and re-enters upload flow).
- ✓ Rejection reason is stored and visible to the provider on their application status page.
- ✓ When status changes (Approved/Rejected/More Docs Needed), the provider receives an email + in-app notification.

Re-application Rules

- ✓ A rejected provider can re-apply after a 7-day cooling period from the rejection date.
- ✓ Re-application resets the review SLA to a fresh 7 days.
- ✓ The provider can see their previous rejection reason and upload corrected documents.
- ✓ Maximum of 3 re-applications. After 3 rejections, the provider must contact support manually.

Document Expiry Rules

- ✓ Documents with expiration dates (insurance, licenses) are tracked by the system.
- ✓ System sends a notification 30 days before a document expires.
- ✓ If a document expires and is not renewed within 14 days, the provider's account is automatically suspended (listings hidden, no new bookings accepted, existing confirmed bookings still honored).
- ✓ Provider can upload a renewed document at any time to lift suspension. Admin must re-approve the new document before the account is reactivated.

1.4 Edge Cases

Edge Cases & Error Handling

- ⚠ User uploads a corrupt or unreadable file → System validates file headers on upload; rejects with message "File could not be read. Please upload a valid PDF, JPG, or PNG."
- ⚠ User tries to submit without all required documents → Submit button is disabled; missing documents are highlighted in red with a tooltip.
- ⚠ Admin does not review within 7 days → System escalates to super-admin with an alert; applicant receives "Your application is taking longer than usual" email.
- ⚠ Provider approved but their MoTA license is revoked externally → No automatic detection. Handled reactively via user reports or periodic manual audits by admin.
- ⚠ Two users try to register as owner of the same business (same license number) → System enforces unique constraint on LicenseNumber per provider type. Second applicant is rejected with "This business is already registered."
- ⚠ Agency removes all affiliated guides after approval → System checks on each guide removal; if count drops to zero, agency status is set to "Suspended" until at least one guide is re-added.

1.5 State Machine

Current State	Action	Next State	Trigger
(none)	User submits application	Pending	User action
Pending	Admin approves	Approved	Admin action
Pending	Admin rejects	Rejected	Admin action
Pending	Admin requests more docs	More Docs Needed	Admin action
More Docs Needed	Provider re-uploads	Pending	User action
Rejected	Provider re-applies (after 7 days)	Pending	User action
Approved	Document expires (14-day grace)	Suspended	System auto
Suspended	Provider uploads renewed doc + admin approves	Approved	Admin action

2. Tour Places & Tour Management

2.1 Place Management

Places are the top-level geographic entities in YallaJo (e.g., Petra, Jerash, Wadi Rum). Only admins can create places. Tours, blogs, and businesses are linked to places.

Place Rules

- ✓ Only admins can create, edit, or delete places.
- ✓ Place name + country must be unique (no duplicate "Petra" entries).
- ✓ Latitude/Longitude are required and validated for range (-90 to 90, -180 to 180).
- ✓ A place cannot be deleted if it has active (non-deleted) tours, businesses, or blogs linked to it. Admin must unlink or archive children first.
- ✓ Places support multi-language translations (Arabic + English at minimum).
- ✓ Accessibility fields (IsWheelchairAccessible, HasAudioGuide, HasBrailleSignage) default to false and are set by admin.

2.2 Tour Creation Workflow

Provider opens dashboard → "Create Tour" → Select Place → Fill details (name, description, difficulty, duration, pricing, meeting point, media) → Set schedule (recurring days + time slots) → Set pricing tiers (if applicable) → Set refund policy → Submit for review (Status: "Pending") → Admin Approve/Reject → If approved: tour goes live and is bookable.

Tour Creation Rules

- ✓ Tour name must be unique per provider (same provider cannot have two tours with the same name).
- ✓ Slug is auto-generated from tour name, must be globally unique.
- ✓ BasePrice must be > 0 and in a supported currency (JOD, USD, EUR).
- ✓ At least 1 media image is required before submission (via EntityImages).
- ✓ MaxGroupSize must be between 1 and 100.
- ✓ DurationMinutes must be between 15 and 2880 (48 hours max for multi-day).
- ✓ Tour must be linked to at least one place (PlaceId required).
- ✓ Tour must have at least one schedule (day + time) before going live.
- ✓ A provider can have a maximum of 50 active tours at any time (prevents spam listings).
- ✓ If the provider's account is suspended, all their tours are automatically hidden from search (but not deleted).

Tour Approval Rules

- ✓ Admin has a 7-day SLA to review new tours (same escalation logic as provider registration).
- ✓ Admin can: Approve, Reject (with reason), or Request Changes.
- ✓ Rejected tours can be edited and re-submitted. No cooling period for tour re-submission.
- ✓ Once approved, edits to critical fields (price, duration, location) set the tour back to "Pending" for re-review. Non-critical edits (description, photos) go live immediately.
- ✓ Critical fields: BasePrice, DurationMinutes, MaxGroupSize, Latitude/Longitude, MeetingPoint.

2.3 Tour Pricing

Pricing Model	How It Works	Example
Flat Rate	Single BasePrice for all participants	25 JOD/person
Tiered Pricing	Price changes based on group size (TourPricingTiers)	1-5 pax: 30 JOD, 6-10: 25 JOD, 11+: 20 JOD
Date-Specific Override	AvailabilitySlot.PriceOverride for peak/off-peak	Friday slots: 35 JOD instead of 25 JOD
Discount Layer	Stacks on top of any pricing model (via Finance.Discounts)	20% off Early Bird applied to tiered price

2.4 Edge Cases

Edge Cases

- ⚠ Provider tries to create a tour for a place that doesn't exist or was deleted → API returns 404; frontend prevents selection of non-active places.
- ⚠ Two providers create tours with the same name for the same place → Allowed (uniqueness is per-provider, not global). Slugs are auto-differentiated with a suffix.
- ⚠ Admin deletes a place that has active tours → Blocked. Admin must archive or reassign tours first.
- ⚠ Provider edits the price of a tour that has confirmed future bookings → Price change applies only to NEW bookings. Existing confirmed bookings retain their original booked price.
- ⚠ Provider tries to set MaxGroupSize lower than the highest BookedCount on any future slot → Blocked with error: "Cannot reduce group size below existing bookings."

3. Categories & Sub-categories

3.1 Workflow

Admin creates the category tree (parent → child hierarchy). Providers select a category when creating a tour or service. Users filter by category during search.

Category Rules

- ✓ Only admins can create, edit, reorder, or delete categories.
- ✓ Maximum nesting depth: 3 levels (e.g., Adventure → Water Sports → Scuba Diving).
- ✓ Category name must be unique within the same parent (no two "Hiking" under "Adventure").
- ✓ Categories support multi-language translations (via CategoryTranslations).
- ✓ A category cannot be deleted if it has active tours or businesses assigned to it. Admin must reassign or archive children first.
- ✓ Deleting a parent category does NOT cascade-delete children. Admin must handle children explicitly.
- ✓ Each tour/business must have at least 1 category and at most 5 categories assigned.
- ✓ Categories have a SortOrder field for controlling display order in the frontend.

Edge Cases

- ⚠ Admin merges two categories (e.g., "Hiking" and "Trekking" into one) → All tours/businesses from the deprecated category are reassigned to the surviving category. Old category is soft-deleted.
- ⚠ Provider selects a category that admin deletes before tour is approved → Tour validation fails on approval; admin is notified to reassign the category.

4. Booking & Reservations System

4.1 Booking Workflow (Tours)

User selects tour → Picks date & time slot → Enters participant count → System checks AvailabilitySlot ($\text{BookedCount} + \text{LockedCount} < \text{MaxCapacity}$) → System creates SlotLock (10-minute TTL) → User proceeds to checkout → Booking created with status "Awaiting Payment" → On payment success: status → "Confirmed", SlotLock released, BookedCount incremented → Confirmation code + receipt sent.

4.2 Reservation Workflow (Businesses)

User selects business → Optionally picks a ServiceItem → Picks date & time → System checks AvailabilitySlot → SlotLock created (10 min) → User proceeds to checkout → Reservation created with status "Awaiting Payment" → On payment: status → "Confirmed".

4.3 Business Rules

Slot & Capacity Rules

- ✓ Slot lock duration: 10 minutes. After expiry, lock is auto-released and the slot becomes available again.
- ✓ Capacity check: $\text{BookedCount} + \text{LockedCount}$ must be $< \text{MaxCapacity}$. Enforced at database level via check constraint.
- ✓ Optimistic concurrency via RowVersion on AvailabilitySlots. If two users try to book the last spot simultaneously, the second one gets a concurrency error and must retry.
- ✓ A user cannot lock the same slot twice. If they abandon checkout and return, the existing lock is reused (not duplicated).
- ✓ SlotLocks are cleaned up by a background job every 5 minutes (releases expired locks and decrements LockedCount).

Booking Rules

- ✓ Maximum 3 concurrent pending (unpaid) bookings per user across all tours/businesses. 4th attempt is blocked with "You have too many pending bookings. Please complete or cancel an existing one."
- ✓ A user cannot book the same tour on the same date twice. Duplicate booking attempt returns an error.
- ✓ Minimum booking lead time: 2 hours before tour start time. No last-second bookings.
- ✓ ParticipantCount must be between 1 and the remaining capacity of the slot.
- ✓ Booking generates a unique ConfirmationCode (format: YJ-YYYYMM-XXXX, e.g., YJ-202603-A7X3) used for check-in.
- ✓ If payment is not completed within 10 minutes (lock duration), the booking is auto-cancelled and the slot is released.
- ✓ Instant booking tours (IsInstantBooking = true) skip provider confirmation and go directly to "Confirmed" on payment. Non-instant tours go to "Pending Confirmation" and the provider has 24 hours to accept or decline.

Provider Confirmation Rules (Non-Instant Booking)

- ✓ After payment, provider receives a notification to accept or decline the booking within 24 hours.
- ✓ If provider does not respond within 24 hours, the booking is auto-confirmed (to protect the user's experience).
- ✓ If provider declines, the user receives a full refund automatically and a notification with the reason.

4.4 Booking State Machine

Current State	Action	Next State
(none)	User submits booking + slot locked	Awaiting Payment
Awaiting Payment	Payment succeeds (instant booking)	Confirmed
Awaiting Payment	Payment succeeds (non-instant)	Pending Confirmation
Awaiting Payment	Lock expires (10 min) / user abandons	Cancelled
Pending Confirmation	Provider accepts	Confirmed
Pending Confirmation	Provider declines	Cancelled + Full Refund

Pending Confirmation	24h passes with no response	Confirmed (auto)
Confirmed	User cancels (refund policy applies)	Cancelled
Confirmed	Tour date passes + completion verified	Completed
Completed	User disputes within escrow period	Disputed
Disputed	Admin resolves	Completed / Refunded

4.5 Edge Cases

Edge Cases

- ⚠ User's payment fails after slot lock → Lock remains for the 10-min duration so user can retry. After 10 min, auto-release.
- ⚠ Two users try to book the last spot simultaneously → First write wins (optimistic concurrency on RowVersion). Loser gets "This slot is no longer available" and is prompted to pick another date.
- ⚠ Provider disables a tour that has future confirmed bookings → Tour is hidden from new searches, but all confirmed bookings are honored. Provider can cancel individual bookings (triggers refund).
- ⚠ User tries to book a tour whose provider is suspended → Tour is hidden from search; if accessed via direct URL, a "This tour is currently unavailable" message is shown.
- ⚠ Network timeout during payment webhook → System implements idempotent webhook handler. Duplicate webhook calls are detected by TransactionId and ignored.
- ⚠ User books 1 spot but wants to add more participants later → Not supported as a modification. User must cancel and rebook with the new count (subject to availability).

5. Payment & Refund Policies

5.1 Payment Workflow

User confirms booking → Redirect to Payment Gateway → Gateway processes payment → Success webhook received → Payment record created (Status: Completed) → Booking status updated to "Confirmed" → Receipt + confirmation code sent via email and in-app.

Payment Rules

- ✓ Supported currency: JOD (primary), USD, EUR. Currency is locked at booking time and cannot change.
- ✓ Payment is processed in full upfront (no partial payments or installments in v1).
- ✓ All monetary amounts stored with decimal(19,4) precision and a companion currency column.
- ✓ Payment gateway response (success/failure/error) is stored in GatewayResponse as JSON for audit.
- ✓ TransactionId from the gateway is stored and used for idempotency checks on webhook retries.
- ✓ Failed payments do NOT create a booking record. The slot lock remains until expiry.
- ✓ System does not store raw credit card numbers. All card data is handled by the payment gateway (PCI DSS compliance).

5.2 Refund Workflow

User requests cancellation → System looks up the RefundPolicy attached to the tour/service → Calculates refund amount based on time-to-tour and policy tiers → Initiates refund via gateway → Payment.RefundedAmount updated → Booking status → "Cancelled" → User notified.

Refund Policy Rules

- ✓ Each provider sets their own refund policy when creating a tour or service.
- ✓ Admin defines global bounds that providers cannot exceed:
 - ✓ — Minimum: Provider must offer at least 50% refund if cancelled 48+ hours before start.
 - ✓ — Maximum: Provider cannot charge more than 100% cancellation fee (no penalties beyond the booking amount).
- ✓ Refund policy has 3 tiers: FullRefundHours (100% refund if cancelled this many hours before), PartialRefundHours (partial refund in this window), and PartialRefundPercent (the percentage refunded in partial window).
- ✓ Below PartialRefundHours: 0% refund (non-refundable).
- ✓ Example: Full refund if cancelled 72h+ before, 50% if 24-72h, 0% under 24h.
- ✓ Refund is always processed back to the original payment method.
- ✓ Refund processing time: instant for wallet/platform credit; 5-14 business days for bank/card (depends on gateway).

- ✓ If a tour is cancelled BY THE PROVIDER, the user always receives a 100% refund regardless of refund policy.
- ✓ If a tour is cancelled due to force majeure (weather, government order), admin can override the refund policy and issue full refunds.

5.3 Payout Workflow

Tour completes (or reservation completed + not refunded) → System waits 7-day escrow period → If no disputes filed → System calculates commission → Initiates transfer to provider's ProviderBankAccount (IsDefault = true) → Generates payout invoice for accounting.

Payout Rules

- ✓ Escrow period: 7 days after tour/reservation completion. During this window, the user can file a dispute.
- ✓ If a dispute is filed during escrow, the payout is frozen until the dispute is resolved by admin.
- ✓ Commission model: Tiered by provider subscription level, with admin override per provider.

Subscription Tier	Default Commission	Admin Can Override?
Free (no subscription)	15%	Yes — per provider
Basic	10%	Yes — per provider
Premium	7%	Yes — per provider
Enterprise	Custom (negotiated)	Yes — per provider

Payout Calculation

- ✓ Net Payout = Booking Amount (after discounts) – Platform Commission.
- ✓ Commission is calculated on the discounted price, not the original price.
- ✓ Example: Tour price 100 JOD, 20% discount applied = 80 JOD paid. Commission (10%) = 8 JOD. Provider receives 72 JOD.
- ✓ Payouts are batched weekly (every Sunday at midnight) for all completed escrow periods.
- ✓ Minimum payout threshold: 10 JOD. If balance is below threshold, it rolls over to the next payout cycle.
- ✓ Provider must have a verified ProviderBankAccount to receive payouts. If no verified account exists, payout is held (not forfeited) until the provider adds one.

- ✓ Each payout generates a PayoutItem per booking for detailed accounting (amount, commission, net).

5.4 Edge Cases

Edge Cases

- ⚠ User requests refund 1 minute before the FullRefundHours cutoff → System uses the exact timestamp comparison. If request is at 71h59m and policy says 72h, it falls into partial refund tier.
- ⚠ Payment gateway is down during refund → Refund is queued with status "Pending" and retried via background job every 15 minutes for up to 24 hours. If still failing, admin is alerted.
- ⚠ Provider changes their refund policy after a user has booked → The policy at the time of booking is the one that applies. The booking stores a snapshot of the policy.
- ⚠ User disputes after the 7-day escrow and payout has already been sent → Late disputes are still accepted but flagged for admin. If admin rules in favor of user, the refund comes from the platform's reserve fund, and the amount is deducted from the provider's next payout.
- ⚠ Provider has no verified bank account when payout is due → Payout is held in escrow indefinitely (not forfeited). Provider gets weekly reminders to add a bank account.
- ⚠ Provider's bank rejects the transfer (wrong IBAN, closed account) → Payout status set to "Failed". Provider notified to update bank details. Retried on next payout cycle after update.

6. Reviews & Ratings

6.1 Review Workflow

Tour/reservation completes → System sends notification: "How was your experience?" → User submits star rating (1-5) + optional title + text + up to 3 photos → System runs profanity filter → Review published on the listing page → Provider notified → AverageRating and ReviewCount on the entity are recalculated.

6.2 Business Rules

Review Eligibility & Submission

- ✓ Anyone can leave a review on any tour or business (no booking required).
- ✓ Reviews from users with a verified/completed booking for that specific tour/business receive a "Verified" badge.
- ✓ Review window: 30 days from tour completion date. After 30 days, the review prompt is no longer shown and the API rejects submissions.
- ✓ Non-verified reviews (no booking) can be submitted at any time but do not receive the badge.
- ✓ A user can only submit ONE review per tour/business. Attempting a second review returns: "You have already reviewed this listing."
- ✓ Star rating is required (1.0 to 5.0 in 0.5 increments). Title and text are optional but encouraged.
- ✓ Maximum review text length: 2000 characters.
- ✓ Maximum 3 photos per review. Each photo max 5 MB. Formats: JPG, PNG.
- ✓ Photos are scanned for NSFW/inappropriate content before publishing (moderation pipeline).

Review Editing & Deletion

- ✓ A user can edit their review within 48 hours of posting. After 48 hours, the review is locked.
- ✓ Edits update the review's UpdatedAt timestamp. Original submission date remains visible.
- ✓ The edit window is based on CreatedAt, not on any subsequent edits (editing does not reset the 48h window).
- ✓ Users can delete their own review at any time (soft delete).
- ✓ Admins can delete any review at any time (for moderation purposes).

Provider Response Rules

- ✓ Providers can reply to any review on their listing with unlimited replies (conversation thread).
- ✓ Provider replies are public and visible to all users on the review thread.
- ✓ Provider replies have no edit window — they can edit or delete their replies at any time.
- ✓ Providers cannot reply to reviews on other providers' listings.
- ✓ Provider replies do NOT affect the star rating calculation.

Rating Calculation

- ✓ AverageRating on the entity is recalculated on every new review, edit, or deletion.
- ✓ Formula: Weighted average where Verified reviews have weight 1.0, non-verified reviews have weight 0.5.
- ✓ Reviews less than 90 days old have a recency boost of 1.0x; reviews older than 90 days get 0.7x weight.
- ✓ Final score is rounded to 2 decimal places (e.g., 4.23).
- ✓ ReviewCount only counts non-deleted reviews.
- ✓ Minimum 3 reviews before AverageRating is displayed publicly. Below 3, the listing shows "Not enough reviews" instead of a numeric score.

6.3 Moderation & Reporting

Report & Moderation Rules

- ✓ Any user can report a review as inappropriate (spam, hate speech, fake, etc.).
- ✓ A user can only report each review once.
- ✓ After 5 unique reports, the review is auto-hidden (not deleted) and flagged for admin review.
- ✓ Admin can: Restore the review (dismiss reports), Permanently delete it, or Warn/ban the reviewer.
- ✓ System runs an automated profanity filter on submission. If triggered, the review is held for manual moderation before publishing (status: "Under Review" instead of "Published").
- ✓ If a user repeatedly has reviews flagged (3+ reviews hidden in 90 days), their future reviews are auto-queued for moderation before publishing.

6.4 Edge Cases

Edge Cases

- ⚠ User submits a review at exactly the 48-hour edit boundary → System uses strict < comparison on UTC timestamps. If CreatedAt + 48h has passed, edit is blocked.
- ⚠ User deletes their review and tries to submit a new one → Allowed. Soft-deleted reviews don't count toward the one-review-per-listing limit.
- ⚠ Provider has 4.9 stars with 100 reviews, then gets 1 spam review → Weighted algorithm minimizes impact. With 100 verified reviews and 1 non-verified review (weight 0.5), the score barely changes. User can also report the spam review.
- ⚠ User uploads a photo that passes format/size checks but contains inappropriate content → Async moderation pipeline flags it post-upload. Photo is hidden until cleared. If the photo was the only one, the review text still publishes.
- ⚠ Provider replies to a review, then the user deletes their review → The entire thread (review + all replies) is soft-deleted together.
- ⚠ User leaves a review for a business they never visited but uses details that suggest they did → No automated way to verify. Review is published without "Verified" badge. Community reporting handles abuse.

7. User Wishlist / Favorites

7.1 Workflow

User clicks "Heart" icon on any tour, place, or business card → Saved to Favorites (EntityType + EntityId)
→ Appears in "My Wishlist" in user profile → If a discount is later activated on a wishlisted item, the user receives a push notification and/or email: "A tour on your wishlist just went on sale!"

7.2 Business Rules

Wishlist Rules

- ✓ Users must be logged in to add to wishlist. Guest users see the heart icon but are prompted to login on click.
- ✓ A user can favorite tours, places, and businesses (via EntityType discriminator).
- ✓ A user can only favorite the same entity once. Clicking the heart again removes it (toggle behavior).
- ✓ No maximum limit on wishlist size.
- ✓ Wishlisted items that are deleted (soft-deleted) by the provider or admin are automatically removed from all users' wishlists on next access (lazy cleanup). Alternatively, a background job cleans up orphaned favorites weekly.
- ✓ Wishlist is private — only the user can see their own wishlist.

Notification Integration (Discount → Wishlist)

- ✓ When a provider activates a discount on a tour or service, the system queries Favorites for all users who wishlisted that entity.
- ✓ Each matching user receives a notification (per their NotificationPreferences: email, push, in-app, or any combination).
- ✓ Notification is sent at most once per discount activation (not on every edit to the discount).
- ✓ If the discount is Subscribers-Only visibility, the notification is only sent to users who have an active subscription.
- ✓ Rate limit: A user receives a maximum of 3 wishlist-discount notifications per day (to avoid spam if many items go on sale simultaneously).

Edge Cases

- ⚠ User wishlists a tour, then the tour is deleted → Favorite remains in DB but is filtered out on read (JOIN finds no matching entity). Background cleanup removes orphans weekly.
- ⚠ Provider activates and deactivates a discount repeatedly → Notification is sent only on the first activation. Subsequent activations within 24 hours are suppressed.
- ⚠ User has notifications disabled but has wishlisted items → No notification sent. The discount badge still appears on the listing card when the user browses.

Phase 2: Experience & Discovery

Business Rules & Edge Cases

8. SEO & Sitemap Enhancement

8.1 Workflow

Background worker runs every 6 hours → Scans all active public entities (tours, places, businesses, blogs) → Generates/updates SitemapEntries table → Renders sitemap.xml endpoint → Pings Google Search Console on changes. Frontend renders dynamic meta tags (Title, Description, OG Image, canonical URL) based on entity data.

8.2 Business Rules

Sitemap Generation Rules

- ✓ Sitemap cron runs every 6 hours (00:00, 06:00, 12:00, 18:00 UTC). Admin can also trigger a manual regeneration on-demand.
- ✓ Only entities with Status = Active/Published and IsDeleted = false are included.
- ✓ Each SitemapEntry stores: URL, ChangeFrequency, Priority, LastModified, EntityType, EntityId.
- ✓ Priority values: Tours = 0.8, Places = 0.9, Businesses = 0.7, Blogs = 0.6, Static pages (About, Contact) = 0.3.
- ✓ ChangeFrequency: Tours = weekly, Places = monthly, Businesses = weekly, Blogs = monthly.
- ✓ LastModified is set to the entity's UpdatedAt timestamp.
- ✓ Maximum 50,000 URLs per sitemap file (Google's limit). If exceeded, the system generates a sitemap index file pointing to multiple sitemap files split by EntityType.
- ✓ After regeneration, the system pings Google Search Console and Bing Webmaster Tools with the sitemap URL.

Meta Tag Rules

- ✓ Every public page must render: <title>, <meta description>, <meta og:title>, <meta og:description>, <meta og:image>, <link rel="canonical">.
- ✓ MetaTitle max length: 60 characters. If not set by admin/provider, auto-generated from entity name.
- ✓ MetaDescription max length: 160 characters. If not set, auto-generated from first 160 chars of the entity's description.
- ✓ OG Image: Uses the entity's primary image (IsPrimary = true from EntityImages). Falls back to the YallaJo default social image if none exists.
- ✓ Canonical URL always points to the slug-based URL (e.g., /tours/petra-by-night) to avoid duplicate content issues.
- ✓ Translated pages use hreflang tags (e.g., hreflang="ar" and hreflang="en") pointing to each other.

Redirect Rules (via `content_seo.Redirects`)

- ✓ When a tour or place slug changes (name edit), the system automatically creates a 301 redirect from the old URL to the new URL.
- ✓ Admins can manually create 301 (permanent) or 302 (temporary) redirects.
- ✓ OldUrl must be unique per active redirect. Duplicate old URLs are rejected.
- ✓ HitCount is incremented on each redirect hit (for analytics).
- ✓ Redirect chains are not allowed: if $A \rightarrow B$ exists and admin tries to create $B \rightarrow C$, the system updates $A \rightarrow C$ directly.
- ✓ Redirects are checked by middleware on every 404 response before returning the error page.

8.3 Edge Cases

Edge Cases

- ⚠ Provider changes tour name (and thus slug) multiple times → Each change creates a new redirect. All old slugs eventually point to the latest slug (chain flattening).
- ⚠ Google Search Console ping fails → Logged as warning. Retried on the next 6-hour cycle. No user-facing impact.
- ⚠ Entity is soft-deleted but still has a SitemapEntry → Sitemap regeneration removes it. Between cycles, the URL may return 404 (acceptable; Google handles this gracefully).
- ⚠ Two entities end up with the same slug due to a race condition → Slug generation appends a random 4-character suffix if uniqueness check fails. Retries up to 3 times.

9. Blog & Tourism Content Management

9.1 Workflow

Admin creates article → Fills title, content (rich text), summary, tags, categories → Optionally links to a Place (PlaceId) → Optionally links related Tours (via BlogTours junction) → Sets status to Draft or Published → If Published, article appears in the Blog section AND as a "Related Read" on the linked place's detail page.

9.2 Business Rules

Blog Authoring Rules

- ✓ Only admins can create, edit, publish, or delete blog articles (v1). Provider-authored blogs are a potential future feature.
- ✓ Blog title max length: 500 characters. Must be non-empty.
- ✓ Slug is auto-generated from title, must be globally unique. Follows the same slug deduplication logic as tours.
- ✓ Summary max length: 1000 characters. Optional but recommended for SEO and listing cards.
- ✓ Content is stored as rich text (nvarchar(max)). Images within the content are stored as external URLs (referencing Attachments).
- ✓ ReadTimeMinutes is auto-calculated: word count / 200 (average reading speed).
- ✓ Each blog can be linked to at most 1 Place (PlaceId) and up to 10 Tours (via BlogTours with SortOrder).
- ✓ Blogs support multi-language translations (via BlogTranslations). Arabic and English are required.

Blog Status Rules

- ✓ Statuses: Draft, Published, Archived.
- ✓ Draft blogs are only visible to admins in the CMS. Not indexed, not public.
- ✓ Published blogs are publicly visible, indexed in the sitemap, and appear in search results.
- ✓ Archived blogs are hidden from the public listing but still accessible via direct URL (for SEO preservation). They display a banner: "This article is archived and may be outdated."
- ✓ PublishedAt is set to the timestamp of the first publish. Subsequent edits update UpdatedAt but not PublishedAt.
- ✓ ViewCount is incremented on each unique page view (debounced: same user viewing within 30 minutes counts as 1 view).

Blog Comments Rules

- ✓ Only logged-in users can comment on blog articles.
- ✓ Comments support nesting (replies) up to 2 levels deep (comment → reply → reply-to-reply; no deeper).
- ✓ Comment max length: 1000 characters.
- ✓ Comments go through the same profanity filter as reviews.
- ✓ Users can react to comments (via BlogCommentReactions). Reaction types: Like, Helpful, Insightful.
- ✓ A user can have only one reaction per comment (changing reaction replaces the previous one).
- ✓ Admins can delete any comment. Users can delete their own comments (soft delete; replies underneath are preserved but show "[deleted]" for the parent).

9.3 Edge Cases

Edge Cases

- ⚠ Admin links a blog to a place, then the place is deleted → Blog's PlaceId is set to null (the blog remains published but is no longer shown as a "Related Read" on any place page).
- ⚠ Admin links a blog to tours, then one of the tours is deleted → BlogTours entry for that tour is removed. Remaining tour links are unaffected.
- ⚠ Blog has comments, then admin archives the blog → Comments are preserved but no new comments can be submitted on archived blogs.
- ⚠ User tries to comment on a draft blog (via direct API call) → API rejects: "Comments are not allowed on unpublished articles."

10. Interactive Map

10.1 Workflow

System fetches entity data (tours, places, businesses) with their Latitude/Longitude → Frontend passes coordinates to Mapbox GL JS → Map renders pins clustered by zoom level → User clicks a cluster to zoom in → User clicks an individual pin → Info popup opens with: entity name, image thumbnail, rating, price (for tours), and a "View Details" link.

10.2 Business Rules

Map Display Rules

- ✓ Map provider: Mapbox GL JS.
- ✓ Default center: Jordan (31.95, 35.93) with zoom level 8 (shows all of Jordan).
- ✓ Pin colors by entity type: Tours = blue, Places = green, Businesses = orange.
- ✓ Clustering is enabled at zoom levels 0–12. At zoom 13+, individual pins are shown.
- ✓ Only active, non-deleted, approved entities with valid coordinates are shown on the map.
- ✓ Coordinates are validated on entity creation: Latitude -90 to 90, Longitude -180 to 180. Invalid coordinates are rejected at the API level.
- ✓ Map data is loaded via a dedicated lightweight API endpoint that returns only: Id, EntityType, Name, Latitude, Longitude, AverageRating, PrimaryImageUrl, BasePrice (for tours). Full entity data is NOT loaded on the map view.
- ✓ Maximum 5,000 pins loaded at a time. If more entities exist, the map uses viewport-based loading (only fetches entities visible in the current bounding box).

Tour Route Display

- ✓ When a user views a specific tour's detail page, the map shows the full tour route using TourWaypoints (ordered by SortOrder).
- ✓ Waypoints are connected by a polyline on the map.
- ✓ The meeting point (MeetingPointLatitude/Longitude) is shown as a distinct pin (flag icon) at the start of the route.
- ✓ Businesses linked to the tour's place (via PlaceBusinesses) are shown as secondary pins on the tour map.

10.3 Edge Cases

Edge Cases

- ⚠ Entity has Latitude/Longitude set to 0.0, 0.0 (likely a data entry error) → Excluded from the map. Admin dashboard shows a "Missing Coordinates" warning for these entities.
- ⚠ Mapbox API rate limit exceeded → Frontend falls back to a cached static map image. User sees a "Map temporarily unavailable" message with a retry button.
- ⚠ User's browser blocks Mapbox scripts (ad blocker) → Map container shows a fallback: list view of entities sorted by distance from the user's location (if geolocation is available) or alphabetically.
- ⚠ Tour has only 1 waypoint → Map shows a single pin instead of a route. No polyline drawn.

11. Weather Forecast Integration

11.1 Workflow

User opens tour detail page → System checks if a cached forecast exists for this location + date range (within 12 hours) → If cache hit: display cached data → If cache miss: API call to weather provider with Lat/Long + date range → Response cached in WeatherCache → Frontend displays forecast (temperature, icon, condition) for the tour's upcoming available dates.

11.2 Business Rules

Weather Rules

- ✓ Weather data is displayed on tour and place detail pages only (not on listing cards).
- ✓ Forecast is shown for the next 7 days from today. Dates beyond 7 days show "Forecast unavailable."
- ✓ Cache TTL: 12 hours per location. After 12 hours, the next request triggers a fresh API call.
- ✓ Cache key: Latitude (rounded to 2 decimals) + Longitude (rounded to 2 decimals) + Date. This means nearby tours at the same place share the same cached weather data.
- ✓ Displayed data: Temperature (Celsius), Weather icon (sunny/cloudy/rainy/etc.), Short condition text (e.g., "Partly Cloudy"), Humidity percentage.
- ✓ Weather provider is abstracted behind an interface (IWeatherProvider) so it can be swapped without code changes. Initial implementation uses a free-tier weather API.
- ✓ API call budget: maximum 1,000 calls per day (free tier limit). Background job pre-fetches weather for the top 50 most-viewed tour locations daily to stay within budget.

11.3 Edge Cases

Edge Cases

- ⚠ Weather API is down or returns an error → Frontend shows "Weather data temporarily unavailable" with a subtle retry link. Stale cache (even if > 12h old) is shown as fallback with a "Last updated X hours ago" label.
- ⚠ Tour location is in a remote area not covered by the weather API → API returns no data. Frontend hides the weather section entirely for that tour.
- ⚠ API rate limit exceeded mid-day → All subsequent requests serve cached data (even if stale). No API calls until the daily limit resets. Admin receives an alert.
- ⚠ User is in a different timezone than the tour location → Weather dates are always shown in the tour location's timezone, not the user's.

12. Smart Notification System

12.1 Workflow

Event triggered (e.g., "Booking Confirmed", "Tour Tomorrow") → System checks NotificationPreferences for the target user → Loads the matching NotificationTemplate → Inserts dynamic data (e.g., user name, tour name, date, confirmation code) → Sends via enabled channels (Email, In-App) → Notification record created with Status = Sent.

12.2 Channels

Channel	Status	Delivery	Notes
In-App (bell icon)	v1 — Active	Real-time via SignalR	Delivered instantly. Persisted in Notifications table.
Email	v1 — Active	Async via background job	Uses NotificationTemplates with HTML layout. Sent via transactional email service.
SMS	v2 — Planned	Async	Reserved for critical alerts (OTP, payment failures). Not in v1.
Push Notification	v3 — Mobile app	Via FCM/APNs	Requires DeviceTokens. Added when mobile app is built.

12.3 Notification Events

Event	Recipients	Channels	When
Booking Confirmed	User	Email + In-App	Immediately on payment success
Booking Cancelled	User + Provider	Email + In-App	Immediately on cancellation
Tour Tomorrow	User	Email + In-App	24 hours before tour start

Tour Completed	User	In-App	When tour status changes to Completed
Review Received	Provider	Email + In-App	When a new review is posted
Review Reply	User (reviewer)	In-App	When provider replies to their review
Provider Application Status	Provider	Email + In-App	On approval/rejection/more docs needed
Document Expiry Warning	Provider	Email + In-App	30 days before document expires
Payout Completed	Provider	Email + In-App	When payout is processed
Dispute Filed	Provider + Admin	Email + In-App	When user files a dispute
Wishlist Sale Alert	User	Email + In-App	When a wishlisted tour goes on discount
Provider Confirmation Needed	Provider	Email + In-App	When non-instant booking needs acceptance
Subscription Expiring	User or Provider	Email + In-App	7 days before subscription renewal

12.4 Business Rules

Notification Delivery Rules

- ✓ In-app notifications are delivered in real-time via SignalR WebSocket connection. If the user is offline, they see unread notifications on next login.
- ✓ Emails are sent asynchronously via a background job queue. Retry policy: 3 attempts with exponential backoff (1 min, 5 min, 15 min).
- ✓ All notifications use templates from NotificationTemplates. Templates support dynamic placeholders: {{UserName}}, {{TourName}}, {{Date}}, {{ConfirmationCode}}, {{Amount}}, etc.
- ✓ Templates are managed by admin via the CMS. Each template has a unique key (e.g., "booking_confirmed"), subject line, HTML body, and plain-text fallback.
- ✓ Templates support multi-language: system selects the template matching the user's preferred language (from UserPreferences). Falls back to English if translation is missing.

User Preference Rules

- ✓ Users can configure notification preferences per event type and per channel.
- ✓ Default: all events enabled for all available channels (Email + In-App).
- ✓ Users CANNOT disable critical notifications: OTP codes, payment confirmations, refund confirmations, and account security alerts. These are always sent.
- ✓ Preference changes take effect immediately (no delay).
- ✓ Unsubscribe link in every email. Clicking it disables email for that specific event type (not all emails).

Notification Lifecycle Rules

- ✓ In-app notifications are auto-marked as read when the user clicks/opens them.
- ✓ Users can bulk "Mark all as read" from the notification panel.
- ✓ Read notifications are auto-deleted after 30 days. Unread notifications are never auto-deleted.
- ✓ Maximum 500 notifications retained per user at any time. If the limit is reached, the oldest read notifications are purged first.
- ✓ Notification bell shows an unread count badge (max display: "99+").

12.5 Edge Cases

Edge Cases

- ⚠ Email service is down → Email is queued and retried 3 times. If all retries fail, the notification is logged as "Failed" and admin is alerted. In-app notification is still delivered independently.
- ⚠ User has no notification preferences record → System uses defaults (all events enabled for all channels). A preferences record is lazily created on first interaction.
- ⚠ Template placeholder has no value (e.g., {{ConfirmationCode}} for a booking that hasn't generated one yet) → Placeholder is replaced with an empty string. Template should be designed to handle this gracefully.
- ⚠ User receives a "Tour Tomorrow" notification but the tour is cancelled between notification send and tour date → A follow-up "Tour Cancelled" notification is sent. The system does not retract already-sent notifications.
- ⚠ SignalR connection drops for a user → Missed in-app notifications are delivered on reconnection (SignalR handles this via persistent connection + message backlog).

13. Children-Friendly Tours

13.1 Workflow

Provider creates/edits tour → Toggles "Suitable for Kids" (IsChildFriendly = true) → Sets recommended MinAge and MaxAge range (e.g., 4–12) → System validates: MinAge must be < MaxAge, both must be between 0 and 17 → Tour card displays "Kids" badge + age range → Users can filter tours by "Family Friendly" in search, and optionally enter their child's age to find matching tours.

13.2 Business Rules

Children-Friendly Rules

- ✓ IsChildFriendly is a boolean toggle on the Tour entity. Default: false.
- ✓ When IsChildFriendly = true, the provider MUST specify MinAge (minimum recommended age). MaxAge (optional, defaults to 17 if not set).
- ✓ MinAge range: 0 (infants welcome) to 17.
- ✓ MaxAge range: MinAge+1 to 17. Must be greater than MinAge.
- ✓ AgeRestriction field (separate from MinAge) is for hard restrictions: if a tour has AgeRestriction = 12, users are warned that children under 12 are NOT ALLOWED (not just "not recommended").
- ✓ If IsChildFriendly = true AND AgeRestriction is set, AgeRestriction must be ≤ MinAge (you can't recommend a tour for 4-year-olds but restrict under-12s).
- ✓ Frontend displays: "Kids" badge on tour card, "Recommended ages: X–Y" on detail page.
- ✓ Search filter: "Family Friendly" checkbox + optional "Child's age" input. When age is entered, only tours where MinAge ≤ child's age ≤ MaxAge are returned.
- ✓ Tours with IsChildFriendly = false are NOT excluded from search when the family filter is off. They simply don't appear when the filter is active.

13.3 Edge Cases

Edge Cases

- ⚠ Provider sets IsChildFriendly = true but doesn't fill MinAge → Validation error: "Please specify the minimum recommended age for children."
- ⚠ Provider sets MinAge = 10 and AgeRestriction = 12 → Validation error: "Age restriction cannot be higher than the minimum recommended age for a children-friendly tour."
- ⚠ Provider marks a "Scuba Diving" tour as kid-friendly with MinAge = 3 → No automatic validation against category (system trusts provider). However, admin can flag this during tour review if the category is typically not suitable for young children.
- ⚠ Parent searches for tours for a 6-year-old. A tour has MinAge=5, MaxAge=10, AgeRestriction=7 → Tour appears in search results (age 6 is in the 5–10 range) but the detail page shows a warning: "Children under 7 are not allowed on this tour."

14. Handicap-Friendly & Accessibility Features

14.1 Workflow

Provider creates/edits tour → Toggles "Accessible" (IsAccessible = true) → Selects applicable accessibility features from a checklist → System saves selections to AccessibilityFeatures table → Tour card displays accessibility icon → Users can filter by accessibility in search → Users with accessibility preferences enabled in their profile see accessible tours prioritized in search results.

14.2 Accessibility Feature Options

Feature	Icon	Description
Wheelchair Accessible		Entire tour route is wheelchair-navigable
Ramps Available		Step-free access via ramps at all stops
Accessible Restrooms		Wheelchair-accessible restrooms available on route
Flat Terrain / No Stairs		No steps, stairs, or steep inclines
Sign Language Guide		Tour guide can communicate in sign language
Audio Guide Available		Pre-recorded audio descriptions available
Braille Signage		Braille labels at key points of interest

14.3 Business Rules

Tour Accessibility Rules

- ✓ IsAccessible is a boolean toggle on the Tour entity. Default: false.
- ✓ When IsAccessible = true, the provider MUST select at least 1 accessibility feature from the checklist.
- ✓ Accessibility features are stored in content_places.AccessibilityFeatures table (FeatureType enum + EntityType + EntityId). This allows both Tours and Places to have accessibility features.

- ✓ Same accessibility features apply to Places (set by admin) and Businesses (set by business owner).
- ✓ Accessibility features from the Place propagate to the tour detail page as "Place Accessibility" alongside the tour's own features.
- ✓ Admin can override or add accessibility features to any tour during the review process.

Search & Discovery Rules

- ✓ Search filter: "Accessible Tours" toggle. When active, only tours with `IsAccessible = true` are shown.
- ✓ Additional sub-filters per accessibility feature (e.g., filter by "Wheelchair Accessible" specifically).
- ✓ Users can set an accessibility preference in their profile (UserPreferences). When set, accessible tours are boosted in search ranking (not exclusive, just prioritized).
- ✓ Accessibility icons are displayed on tour cards, place pages, and business listings.
- ✓ The "Smart Accessibility" feature (Phase 4, Feature 23) adds UI-level adaptations: high contrast mode, screen reader ARIA enrichment, and font size scaling. Those are frontend-only and do not interact with this data model.

Accessibility Review Integration

- ✓ Users can submit AccessibilityReviews (via `social.AccessibilityReviews` table) to rate how accurate the accessibility claims are.
- ✓ AccessibilityReview is separate from the main review. It asks: "Were the accessibility features accurate?" (Yes/No) + optional comment.
- ✓ Only users who completed a booking for that tour can submit an accessibility review.
- ✓ If more than 3 accessibility reviews report inaccurate claims, the tour is flagged for admin review and a warning badge is shown: "Accessibility features unverified."

14.4 Edge Cases

Edge Cases

- ⚠ Provider marks tour as accessible but selects no features → Validation error: "Please select at least one accessibility feature."
- ⚠ Provider claims "Wheelchair Accessible" but the linked Place is not wheelchair accessible → System shows a mismatch warning to admin during tour review: "Tour claims wheelchair access but linked place does not." Admin decides whether to approve.
- ⚠ A place updates its accessibility features (e.g., adds ramps) → Tours linked to that place do NOT auto-update. Provider must update their own tour's accessibility claims. A notification is sent to the provider: "The accessibility features of [Place] have been updated. Please review your tour's claims."

⚠ User with accessibility preferences searches but no accessible tours exist in their area → Search returns all tours (no accessible filter forced) with a message: "No accessible tours found in this area. Showing all available tours."

Phase 3: Growth & Monetization

Business Rules & Edge Cases

15. Packaging System

15.1 Workflow

Admin opens "Packages" in CMS → Creates a new TourPackage: selects a base Tour + adds TourPackageInclusions (linked businesses: hotel, restaurant, activity) → Sets bundle price (either a fixed % off the combined price OR a custom flat price) → Sets validity window (ValidFrom/ValidTo) and MaxParticipants → Package goes live → User books package → System reserves slots in the tour AND all included businesses atomically → Single payment for the entire package.

15.2 Business Rules

Package Creation Rules

- ✓ Only admins can create, edit, or deactivate packages.
- ✓ Minimum 2 items in a package (e.g., 1 tour + 1 hotel). Maximum 10 items.
- ✓ Every package must include exactly 1 tour (the anchor). Additional items are businesses (hotels, restaurants, cafes, activities).
- ✓ Bundle pricing supports two modes:
 - ✓ — Percentage discount: Admin sets a discount % off the sum of individual prices (e.g., 15% off). System auto-calculates the package price.
 - ✓ — Custom flat price: Admin sets any price, but it must be lower than the sum of individual prices (no upcharging via packages).
- ✓ Package price is stored on TourPackages.Price. It is recalculated if any included item's price changes (admin is notified to review).
- ✓ ValidFrom / ValidTo define when the package is bookable. Outside this window, the package is hidden from search.
- ✓ MaxParticipants on the package is the minimum of all included items' capacities (bottleneck determines the limit).
- ✓ Packages can be marked IsActive = false to hide them without deleting.

Package Booking Rules

- ✓ When a user books a package, the system must atomically reserve availability slots for the tour AND all included businesses.
- ✓ If ANY item in the package has no availability for the selected date, the entire package booking fails. No partial bookings.
- ✓ The 10-minute slot lock applies to ALL items simultaneously. If the user abandons checkout, all locks are released together.
- ✓ A single Payment record is created for the full package price. A PackageBookings record links the payment to the package.
- ✓ The system creates individual TourBooking and Reservation records under the hood for each item, all linked to the same PackageBookings.Id.

- ✓ Cancellation of a package cancels ALL items. Partial cancellation (keeping the hotel but cancelling the tour) is not supported in v1.
- ✓ Refund policy for packages: the MOST RESTRICTIVE refund policy among all included items applies to the entire package.

Payout Distribution Rules

- ✓ On package completion, the platform distributes payouts to each provider proportionally.
- ✓ Distribution formula: each provider's share = (their item's individual price / sum of all individual prices) × package price paid.
- ✓ Commission is deducted per provider based on their individual subscription tier.
- ✓ Example: Package price 150 JOD. Tour (100 JOD individual) + Hotel (80 JOD individual). Sum = 180 JOD. Tour provider share = $(100/180) \times 150 = 83.33$ JOD. Hotel share = $(80/180) \times 150 = 66.67$ JOD. Commission deducted from each individually.

15.3 Edge Cases

Edge Cases

- ⚠ A business included in a package is deactivated or deleted → Package is auto-deactivated (IsActive = false). Admin is notified: "Package [X] deactivated because [Business Y] is no longer available." Existing confirmed bookings are honored.
- ⚠ Provider of an included business changes their price → Package price is NOT auto-updated. Admin receives a notification: "The price of [Business Y] has changed. Please review Package [X]." Admin decides whether to adjust the bundle price.
- ⚠ User tries to book a package but one business has no availability on the selected date → Booking fails with: "[Business Y] is not available on this date. Please select a different date." All slot locks are released.
- ⚠ Package has ValidTo in the past → Package is automatically hidden from search by the sitemap cron. Direct URL access shows: "This package is no longer available."
- ⚠ Two users try to book the last-available package simultaneously → Same optimistic concurrency as individual bookings. First to complete payment wins. Second gets availability error.

16. Paid Features & Subscription Plans

16.1 Provider Subscriptions

Provider visits "Upgrade" page → Views available tiers (Free, Basic, Premium, Enterprise) → Selects a plan and billing cycle (Monthly or Annual) → Annual plans show a discount (e.g., "Save 20%") → Pays via payment gateway → System activates subscription → Provider flag updated → Benefits applied immediately.

Tier	Monthly Price	Annual Price	Commission	Key Benefits
Free	0 JOD	0 JOD	15%	Basic listing, standard search ranking
Basic	15 JOD	144 JOD (save 20%)	10%	Priority search boost, analytics dashboard, featured badge
Premium	35 JOD	336 JOD (save 20%)	7%	Top search ranking, homepage featured section, advanced analytics, priority support
Enterprise	Custom	Custom	Negotiated	Dedicated account manager, custom integrations, API access, bulk management tools

Provider Subscription Rules

- ✓ 30-day free trial available for Basic and Premium tiers. Trial starts on first subscription activation. One trial per provider account (ever).
- ✓ Trial does not require payment method upfront. 3 days before trial ends, provider receives a reminder to add payment and subscribe.
- ✓ If trial expires without subscribing, provider reverts to Free tier. All premium features are removed but listings remain active.
- ✓ Billing cycles: Monthly (auto-renews on the same day each month) or Annual (auto-renews yearly).
- ✓ Annual plans are discounted at 20% off the equivalent monthly total.

- ✓ Subscription payment fails on renewal → System retries 3 times over 7 days (day 1, day 3, day 7). During this grace period, premium features remain active. If all retries fail, subscription downgrades to Free.
- ✓ Provider can upgrade tier at any time. The price difference is prorated for the remaining billing period.
- ✓ Provider can downgrade at any time. Downgrade takes effect at the end of the current billing period (no mid-cycle downgrades).
- ✓ Provider can cancel subscription at any time. Access continues until the end of the paid period. No refunds for unused time.
- ✓ Enterprise tier requires manual approval by admin. Provider submits an inquiry, admin sets custom pricing.

16.2 User Subscriptions

User visits "YallaJo+" page → Selects a tier → Pays → System updates user flag → Benefits applied.

User Subscription Benefits

- ✓ Subscriber-only discounts: users see exclusive discounts not visible to free users.
- ✓ Priority reservations: during high-demand periods, subscribers' slot locks are extended to 15 minutes (vs. 10 for free users).
- ✓ Best deal notifications: subscribers receive real-time alerts for new deals before they're shown publicly (24-hour early access).
- ✓ Loyalty point boost: subscribers earn 1.5x loyalty points on every booking.
- ✓ Same billing mechanics as provider subscriptions (monthly/annual, auto-renew, grace period, cancellation).

16.3 Subscription Features System

Feature Gating Architecture

- ✓ SubscriptionFeatures table defines available features (e.g., "analytics_dashboard", "search_boost", "priority_support").
- ✓ PlanFeatures junction table links features to plans with an optional Value (e.g., search_boost: "1.5x" for Basic, "3x" for Premium).
- ✓ Feature checks are performed via a centralized SubscriptionService that caches the current user's plan and features.
- ✓ Admin can add new features to the SubscriptionFeatures table and assign them to plans without code changes.
- ✓ If a feature is removed from a plan, existing subscribers on that plan lose access at their next billing cycle (not immediately).

16.4 Edge Cases

Edge Cases

- ⚠ Provider on Premium tier gets their account suspended (expired documents) → Subscription billing continues (they still owe for the plan), but premium features are inactive until suspension is lifted. Provider can cancel subscription during suspension.
- ⚠ Provider upgrades from Basic (monthly) to Premium (annual) mid-cycle → System calculates: unused days on Basic are credited. Premium annual price minus credit = amount charged. New annual cycle starts from upgrade date.
- ⚠ Payment gateway is down during renewal → Retry schedule (day 1, 3, 7) with exponential backoff. Premium features remain during grace period. If all retries fail, downgrade + notification.
- ⚠ Admin changes the price of a plan → Existing subscribers keep their current price until their next renewal. New price applies only to new subscribers and at renewal time.
- ⚠ Provider uses free trial, cancels, then tries to get another trial → System checks: trial already used (stored on Subscriptions history). Trial option is not shown; only paid plans are available.

17. Referral & Loyalty System

17.1 Referral Workflow

User A opens "Invite Friends" page → Copies unique referral link (contains ReferralCode) → Shares with User B → User B signs up via the link → System records linkage (Referrals table: ReferrerUserId, ReferredUserId) → User B completes their first booking → System credits points to User A (referrer reward) AND applies a discount to User B's first booking (referred reward).

17.2 Referral Rules

Referral Rules

- ✓ Each user gets one unique ReferralCode (auto-generated, format: YJ-XXXXXX). It never changes.
- ✓ Referral link: <https://yallajo.com/join?ref=YJ-XXXXXX>
- ✓ The referred user (User B) must sign up via the link AND complete their first booking for the referral to be "Completed."
- ✓ Referrer reward: loyalty points credited to User A's LoyaltyPoints balance. Amount is admin-configurable (default: 100 points).
- ✓ Referred reward: a one-time discount on User B's first booking. Amount is admin-configurable (default: 5 JOD off, stored as ReferredRewardAmount on the Referral record).
- ✓ The referred discount is auto-applied at checkout for User B's first booking. It cannot be stacked with other promo codes but CAN stack with public/automatic discounts.
- ✓ A user cannot refer themselves (same email/phone detection).
- ✓ A user can only be referred once. If they already have an account, the referral link has no effect.
- ✓ No limit on how many people a user can refer.
- ✓ Referral status flow: Pending (signed up, no booking) → Completed (first booking paid) → rewards distributed.
- ✓ If User B's first booking is later refunded, the referral remains Completed and rewards are NOT clawed back (to keep the system simple and trust-based).

17.3 Loyalty Points System

Points Earning Rules

- ✓ Points are earned via specific actions. Earning rates are admin-configurable:

Action	Default Points	Conditions
Complete a tour booking	10 points per 1 JOD spent	Awarded after tour completion (not on booking)

Complete a reservation	10 points per 1 JOD spent	Awarded after reservation completion
Write a verified review	25 points	Must be for a completed booking
Refer a friend (completed)	100 points	Awarded when referred user completes first booking
First booking ever	50 bonus points	One-time welcome bonus
Birthday bonus	100 points	Credited on user's birthday (if birthdate is on profile)

Points Redemption Rules

- ✓ Points can be redeemed for discounts on bookings and subscription payments.
- ✓ Conversion rate is admin-configurable. Default: 100 points = 1 JOD.
- ✓ Minimum redemption: 100 points (1 JOD equivalent at default rate).
- ✓ Maximum redemption per booking: 50% of the booking total. Users cannot pay 100% with points.
- ✓ At checkout, user chooses how many points to redeem (slider or input). System shows: "Redeem 500 points for 5 JOD discount."
- ✓ Redeemed points are deducted from AvailablePoints immediately on payment. If payment fails, points are refunded.
- ✓ Points redeemed for a booking that is later cancelled: points are refunded back to the user's balance.
- ✓ Subscribers earn 1.5x points on all actions (e.g., 15 points per JOD instead of 10).

Points Expiry & Balance Rules

- ✓ Points expire 12 months after the date they were earned (per-transaction expiry, not bulk).
- ✓ Expiry is tracked via LoyaltyTransactions.ExpiresAt (set to CreatedAt + 12 months on earning).
- ✓ A background job runs daily to expire old points: sets expired transactions to inactive and decrements AvailablePoints on LoyaltyPoints.
- ✓ System notifies users 30 days before their oldest points batch is about to expire: "You have 200 points expiring on [date]. Use them before they expire!"
- ✓ LoyaltyPoints table maintains 3 balances: TotalPoints (lifetime earned), AvailablePoints (current redeemable), LifetimePoints (historical total for tier tracking).
- ✓ When redeeming, the system uses FIFO: oldest points (closest to expiry) are consumed first.

17.4 Edge Cases

Edge Cases

- ⚠ User earns points for a booking, then the booking is disputed and refunded → Points awarded for that booking are clawed back (deducted from AvailablePoints). If AvailablePoints goes negative (user already spent them), the negative balance carries forward and is deducted from future earnings.
- ⚠ User has 99 points and tries to redeem → Blocked: "Minimum redemption is 100 points. You need 1 more point."
- ⚠ User tries to redeem more points than 50% of booking total → Slider/input is capped at 50%. System shows: "Maximum point discount for this booking is X JOD."
- ⚠ User's points expire while they're in the middle of checkout → Expiry job skips points that are currently locked in an active SlotLock. Points expire after the lock releases (if unused).
- ⚠ Admin changes conversion rate from 100:1 to 50:1 → Change applies immediately to all future redemptions. Existing earned points retain their value at the new rate (points become more valuable). No retroactive adjustments.
- ⚠ Referrer creates fake accounts to farm points → Mitigation: referred user must complete a PAID booking (not just sign up). Same-device/IP detection flags suspicious referrals for admin review.

18. Rating Algorithm

18.1 Workflow

New review submitted → System recalculates the entity's AverageRating using a weighted algorithm → Updates AverageRating and ReviewCount on the entity (Tour, Place, Business, TourGuide) → Updated score affects search ranking and listing display.

18.2 Algorithm Details

Weighted Rating Formula

- ✓ Each review receives a composite weight = Verification Weight × Recency Weight.
- ✓ Verification Weight: Verified purchase (completed booking) = 1.0, Non-verified = 0.5.
- ✓ Recency Weight: Review less than 90 days old = 1.0, 90–180 days = 0.7, older than 180 days = 0.5.
- ✓ $\text{AverageRating} = \text{Sum}(\text{rating}_i \times \text{weight}_i) / \text{Sum}(\text{weight}_i)$ for all non-deleted reviews.
- ✓ Result is rounded to 2 decimal places.
- ✓ ReviewCount = total number of non-deleted reviews (unweighted count).

Display & Ranking Rules

- ✓ Minimum 3 reviews required before AverageRating is displayed. Below 3: "Not enough reviews" shown.
- ✓ Entities with 0 reviews appear last in search results (lowest ranking).
- ✓ Search ranking incorporates AverageRating with a Bayesian prior: entities with few reviews are pulled toward the global average to prevent a single 5-star review from dominating rankings.
- ✓ Bayesian formula: $\text{BayesianScore} = (\text{ReviewCount} \times \text{AverageRating} + C \times \text{GlobalAverage}) / (\text{ReviewCount} + C)$, where $C = 10$ (confidence threshold, admin-configurable).
- ✓ AverageRating is recalculated on: new review, review edit, review deletion, and daily by a background job (to account for recency decay).
- ✓ The daily recalculation job runs at 03:00 UTC and processes all entities with at least 1 review.

18.3 Rating Impact on Visibility

AverageRating Range	Effect
4.5 – 5.0	Eligible for "Top Rated" badge and homepage featured section
4.0 – 4.49	Standard ranking, appears in "Popular" sections

3.0 – 3.99	Standard ranking, no special treatment
2.0 – 2.99	Deprioritized in search. Provider receives a nudge: "Your rating is below average."
Below 2.0	Admin flagged for review. Provider receives a warning. If sustained for 90 days, listing may be suspended.

18.4 Edge Cases

Edge Cases

- ⚠ Entity has exactly 3 reviews: one 5-star (verified, recent), one 1-star (non-verified, old), one 3-star (verified, recent) → Weights: $(5 \times 1.0 \times 1.0) + (1 \times 0.5 \times 0.5) + (3 \times 1.0 \times 1.0) = 5 + 0.25 + 3 = 8.25$. Sum of weights: $1.0 + 0.25 + 1.0 = 2.25$. Score: $8.25 / 2.25 = 3.67$.
- ⚠ All reviews are deleted (ReviewCount drops to 0) → AverageRating resets to 0.00. Display shows "Not enough reviews."
- ⚠ Competitor leaves fake negative reviews → Non-verified reviews have half weight. Combined with reporting (auto-hide after 5 reports) and admin moderation, impact is minimized.
- ⚠ Provider has 4.9 rating and receives a flood of 1-star non-verified reviews → Each has weight 0.5 (non-verified) $\times 1.0$ (recent) = 0.5 . Against 50 verified reviews (weight 1.0 each), the impact is significantly diluted. Admin is alerted if ReviewCount spikes abnormally (>10 reviews in 24h).

19. Dispute Center

19.1 Workflow

User reports issue (e.g., "Guide didn't show up") → Creates a Dispute linked to the Payment → Booking payout is frozen (if still in escrow) → Admin dashboard opens ticket → Both user and provider can submit messages (DisputeMessages) and upload evidence (DisputeEvidence: photos, receipts, screenshots) → Admin reviews all evidence → Admin resolves: Full Refund / Partial Refund / Platform Credit / Reject Claim.

19.2 Business Rules

Filing Rules

- ✓ Users can file a dispute within 7 days of tour/reservation completion (aligned with escrow period).
- ✓ A dispute can only be filed for bookings with status "Completed" or "Confirmed" (post-tour).
- ✓ One dispute per booking. If a dispute is rejected, the user cannot file another for the same booking.
- ✓ Required fields: Reason (selected from predefined list), Description (free text, max 2000 chars).
- ✓ Predefined dispute reasons: Guide no-show, Service not as described, Safety concern, Overcharged, Tour cancelled without notice, Poor quality, Other.
- ✓ Upon filing, the booking's payout (if still in escrow) is immediately frozen. If payout has already been processed (late dispute), it's flagged for admin but not auto-reversed.

Evidence & Communication Rules

- ✓ Both the user and the provider can submit messages via DisputeMessages (visible to both parties + admin).
- ✓ Admin can post internal-only messages (IsInternal = true) visible only to admin team, not to user or provider.
- ✓ Both parties can upload evidence via DisputeEvidence: screenshots, receipts, photos, documents.
- ✓ Max 10 evidence files per party per dispute. Max 5 MB per file. Formats: PDF, JPG, PNG.
- ✓ Provider is notified immediately when a dispute is filed and has 48 hours to respond with their side. If they don't respond within 48h, admin may proceed with the user's evidence only.
- ✓ Admin has a 7-day SLA to resolve disputes from the filing date.

19.3 Resolution Options

Resolution	What Happens	Who Pays
Full Refund	100% of booking amount refunded to user's original payment method	Deducted from provider's pending payout or next payout
Partial Refund	Admin sets % (e.g., 50%). Refunded to user's original payment method	Deducted from provider's payout proportionally
Platform Credit	Amount credited to user's loyalty points or platform wallet (future use only)	Platform absorbs the cost (goodwill gesture)
Reject Claim	No refund. Payout released to provider	No financial impact

Resolution Rules

- ✓ Admin must select a resolution AND write resolution notes (ResolutionNotes, max 2000 chars) explaining the decision.
- ✓ Both user and provider are notified of the resolution via email + in-app notification.
- ✓ Full/Partial Refund: refund is initiated immediately via payment gateway. If the original payment method is unavailable, platform credit is issued instead.
- ✓ Platform Credit: converted to loyalty points at the current admin-configured conversion rate. User can use them on future bookings.
- ✓ After resolution, the dispute status is set to "Resolved" and no further messages or evidence can be submitted.
- ✓ If payout was already processed before the dispute was filed (late dispute within 7 days but after payout batch ran), the refund amount is deducted from the provider's next payout cycle. If provider has no future payouts, the amount is recorded as a debt and deducted when they do.

19.4 Dispute State Machine

Current State	Action	Next State
(none)	User files dispute	Open
Open	Provider responds within 48h	Under Review
Open	Provider does not respond in 48h	Under Review (admin proceeds)

Under Review	Admin requests more evidence	Awaiting Evidence
Awaiting Evidence	Party submits evidence	Under Review
Under Review	Admin resolves: refund	Resolved — Refunded
Under Review	Admin resolves: credit	Resolved — Credited
Under Review	Admin resolves: reject	Resolved — Rejected

19.5 Edge Cases

Edge Cases

- ⚠ User files dispute on day 7 at 23:59 UTC → Accepted (strict ≤ 7 days check). Filing at day 8 00:00 is rejected.
- ⚠ User files dispute but payout was already sent to provider → Dispute is accepted. If admin rules in user's favor, refund comes from platform reserve fund. Amount is deducted from provider's next payout as a "chargeback."
- ⚠ Provider has no future bookings (and thus no future payouts) to deduct from → Debt is recorded. If provider reactivates and gets new bookings, the debt is deducted from the first payout. If the provider's account is permanently closed, the debt is written off by the platform.
- ⚠ Admin takes longer than 7 days to resolve → System sends escalation alerts at day 5 and day 7 to super-admin. User receives: "Your dispute is being reviewed and is taking longer than expected."
- ⚠ User uploads evidence that contains personally identifiable information of a third party → Admin can flag and redact evidence files. The original file is preserved for legal records but a redacted version is shown to the other party.
- ⚠ Both user and provider provide contradictory evidence with no clear resolution → Admin can issue a compromise: partial refund or platform credit as a goodwill gesture, with resolution notes explaining the decision.

Phase 4: Advanced Features & AI

Business Rules & Edge Cases

20. Live Tour & Tour Guide Tracking

20.1 Workflow

Tour start time arrives → Guide opens their dashboard and taps "Start Tour" → System creates a LiveTrackingSession (Status: Active) → Guide's device begins broadcasting GPS coordinates to the server every 30 seconds via SignalR WebSocket → Server stores each snapshot in LocationSnapshots → Booked participants and admin subscribe to the tour's tracking channel → Map on their screen updates the guide's pin in real-time → Guide reaches TourCheckpoints (predefined waypoints) → System auto-detects proximity and logs arrival → Tour ends → Guide taps "End Tour" → Session status: Completed → Full route is available for replay.

20.2 Business Rules

Session Lifecycle Rules

- ✓ A LiveTrackingSession can only be started by the assigned tour guide for a confirmed booking on the scheduled date.
- ✓ Guide can start the session up to 30 minutes before the scheduled start time (early arrival) and no later than 30 minutes after (late start grace period).
- ✓ If the guide does not start the session within 30 minutes of the scheduled start, all booked participants receive a notification: "Your tour guide has not started tracking. Please contact support if the guide has not arrived."
- ✓ Only one active session per guide at a time. Starting a new session while one is active is blocked.
- ✓ Session auto-ends after the tour's DurationMinutes + 60-minute buffer (safety margin for delays). Guide can also end it manually at any time.
- ✓ Session statuses: Scheduled → Active → Completed / Abandoned.
- ✓ If the guide's app crashes or loses connection for more than 10 minutes without any GPS data, the session status changes to "Connection Lost" (sub-status of Active). Participants see: "Guide's connection lost. Last known location shown."

GPS Broadcasting Rules

- ✓ Broadcast interval: every 30 seconds.
- ✓ Each LocationSnapshot stores: Latitude, Longitude, Altitude (optional), Accuracy (meters), Speed (optional), RecordedAt timestamp, SessionId.
- ✓ Minimum accuracy threshold: snapshots with accuracy > 100 meters are stored but marked as low-quality and not displayed on the live map (prevents indoor/tunnel GPS drift from confusing users).
- ✓ Server validates incoming coordinates: Latitude (-90 to 90), Longitude (-180 to 180). Invalid data is silently dropped.
- ✓ Data is broadcast to subscribers via SignalR hub. Only authenticated users who are booked on this specific tour can subscribe to the channel.

- ✓ Admin can subscribe to any active tour's tracking channel for monitoring purposes.
- ✓ No public/unauthenticated access to live tracking data.

Offline Mode Rules

- ✓ When the guide's device loses internet connectivity, the app continues recording GPS snapshots locally in device storage.
- ✓ Local storage buffer: up to 2 hours of snapshots (240 snapshots at 30-second intervals).
- ✓ When connectivity is restored, the app uploads all buffered snapshots in a single batch, sorted by RecordedAt timestamp.
- ✓ Server processes the batch and fills in the gap on subscribers' maps (they see the route "draw in" for the missed period).
- ✓ If the buffer exceeds 2 hours, the oldest snapshots are discarded (FIFO) to prevent excessive memory usage on the device.
- ✓ During offline periods, subscribers see the guide's last known location with a "Last updated X minutes ago" label.

Checkpoint Rules

- ✓ TourCheckpoints are predefined by the tour guide during tour creation (linked to TourWaypoints).
- ✓ Auto-detection: when the guide's GPS is within 50 meters of a checkpoint for at least 30 seconds, the checkpoint is automatically marked as "Reached" with a timestamp.
- ✓ Participants see checkpoint progress on their map: completed checkpoints turn green, upcoming ones stay gray.
- ✓ If a guide skips a checkpoint (route deviation), it remains gray. No penalty, but the data is logged for analytics.
- ✓ Admin can view checkpoint completion rates per tour for quality monitoring.

20.3 Data Retention & Privacy

Privacy & Retention Rules

- ✓ Live tracking data (LocationSnapshots) is retained for 30 days after the tour ends, then auto-purged by a background job.
- ✓ During the 30-day retention window, the data is available for: dispute evidence, route replay by admin, and quality analytics.
- ✓ After 30 days, only aggregated data is kept: total distance, average speed, checkpoint completion (no individual GPS points).
- ✓ Guide's live location is NEVER shown when they are not in an active session. There is no passive tracking.
- ✓ Participants can only see the guide's location during the active session window. After the session ends, live tracking stops immediately.

- ✓ GDPR/privacy: guide consents to tracking when they accept a booking. Consent is recorded on the TourGuide entity.

20.4 Edge Cases

Edge Cases

- ⚠ Guide starts the tour but forgets to tap "Start Tour" → Participants see no tracking. After 30 minutes post-scheduled-start, a reminder notification is sent to the guide: "Did you forget to start tracking?"
- ⚠ Guide's phone battery dies mid-tour → Session remains Active with "Connection Lost" status. Offline buffer (if any) uploads when the guide gets another device or charges. If no data arrives for the rest of the tour, the session is auto-ended after DurationMinutes + 60 min buffer.
- ⚠ Two participants from the same booking check the map simultaneously → Both receive the same SignalR broadcast. No performance issue — they share the same channel.
- ⚠ Guide is in an area with GPS spoofing or drift (tunnels, indoor sections) → Low-accuracy snapshots (>100m) are hidden from the live map. When the guide emerges, accurate snapshots resume and the route continues from the last good point.
- ⚠ Tour is cancelled after the tracking session has started → Admin or guide can force-end the session. Status: Abandoned. Participants are notified. Location data for the partial session is retained for dispute purposes.
- ⚠ Multiple tours running simultaneously for the same guide (shouldn't happen) → System blocks: only one active session per guide. If a scheduling conflict exists, it should have been caught at booking time.

21. Recommendation System

21.1 Workflow

User browses the site (views tours, clicks listings, searches, books, reviews) → System logs all interactions in UserInteractions → Daily background job runs the recommendation engine → Hybrid model generates a ranked list of recommended tours per user → Results cached in RecommendationCaches → On page load, real-time boost layer re-ranks based on the user's current session activity → "Recommended for You" section displayed on homepage and tour listing pages.

21.2 Recommendation Approach

Hybrid Model: Three Signals

- ✓ Signal 1 — Collaborative Filtering (weight: 40%): Users who booked/viewed the same tours as you also liked these tours. Based on user-item interaction matrix. Minimum 5 interactions required before collaborative filtering kicks in; below that, fallback to content-based + popularity only.
- ✓ Signal 2 — Content-Based (weight: 35%): Match the user's implicit preferences (derived from viewed/booked tour attributes: categories, difficulty, duration, price range, location) against all available tours. Uses cosine similarity on a feature vector.
- ✓ Signal 3 — Popularity (weight: 25%): Global popularity score from PopularityScores table (based on BookingCount, ViewCount, AverageRating, recency). Acts as a fallback for cold-start users and ensures trending tours surface.
- ✓ Final score = $0.4 \times \text{CollaborativeScore} + 0.35 \times \text{ContentScore} + 0.25 \times \text{PopularityScore}$.
- ✓ Weights are admin-configurable in system settings.

Data Collection Rules

- ✓ UserInteractions table logs: InteractionType (View, Click, Search, Bookmark, Book, Review), EntityType, EntityId, Timestamp, SessionId.
- ✓ Interactions are logged for both logged-in users (linked to UserId) and guests (linked to SessionId via cookie). Guest data is merged into the user's profile on login/signup.
- ✓ Search queries are stored (SearchKeyword field) for content-based signal extraction.
- ✓ Only the last 90 days of interactions are used for model training. Older data is archived.
- ✓ Minimum interactions for personalized recommendations: 5 views or 1 booking. Below this threshold, the user sees popularity-only recommendations.

Caching & Refresh Rules

- ✓ RecommendationCaches stores: UserId, RecommendedEntityId, Score, Rank, GeneratedAt.
- ✓ Daily batch job runs at 02:00 UTC: recalculates recommendations for all users with at least 5 interactions in the last 90 days.

- ✓ Cache stores top 50 recommendations per user.
- ✓ Real-time boost layer: when a user views or interacts with something during their current session, the cached recommendations are re-ranked on-the-fly to reflect this (e.g., if user just viewed a Petra tour, other Petra tours get a temporary boost).
- ✓ Cache TTL: 24 hours. If the user requests recommendations and the cache is stale, the popularity-based fallback is served while the batch job catches up.
- ✓ For users with no cache (new users, guests), popularity-only recommendations are served instantly (no computation delay).

21.3 Display Rules

Recommendation Display Rules

- ✓ "Recommended for You" section: shows top 10 tours on the homepage, personalized per user.
- ✓ "Similar Tours" section: on each tour detail page, shows 6 related tours (content-based similarity to the current tour).
- ✓ "Trending Now" section: shows top 10 tours by PopularityScore globally (not personalized). Refreshed every 6 hours.
- ✓ "Recently Viewed": shows the user's last 10 viewed tours (from UserInteractions, not the model). Strictly client-side augmented with server data.
- ✓ Recommendations NEVER include: tours the user has already booked, tours from suspended providers, inactive/deleted tours, or tours in locations the user has explicitly hidden (future preference feature).
- ✓ Diversity rule: no more than 3 tours from the same provider in a single recommendation list (prevents one provider dominating).

21.4 Edge Cases

Edge Cases

- ⚠ Brand new user with zero interactions → Popularity-only recommendations (Trending Now = Recommended for You). After 5 interactions, personalized recommendations begin appearing within 24 hours (next batch run).
- ⚠ User only books one category (e.g., always "Hiking") → Content-based signal would only recommend hiking. The popularity signal (25%) ensures some diversity by surfacing highly-rated tours from other categories. A future "exploration bonus" could further boost diversity.
- ⚠ Two users with identical interaction histories → Collaborative filtering gives them similar recommendations, but content-based and real-time boost will diverge as their sessions differ.
- ⚠ Tour that was popular last month but has no recent bookings → Popularity score decays over time (recency factor). It gradually drops out of Trending and recommendations unless it continues to get views.
- ⚠ Provider creates 50 tours to game the recommendation system → Diversity rule (max 3 per provider per list) limits exposure. Additionally, tours with zero bookings and low views have near-zero popularity scores.

⚠ Recommendation batch job fails or times out → Stale cache is served (up to 48h old). If cache is completely empty, popularity fallback is used. Admin is alerted on job failure.

22. AI Assistant & Chatbot

22.1 Workflow

User clicks the chat icon (floating button, bottom-right) → Chat window opens → User types a question in English or Arabic (e.g., "Best beach tour for kids?") → System sends the message to the AI layer (external API in v1) with context: user preferences, location, booking history (if logged in), and the YallaJo tour database → AI generates a natural language response with specific tour recommendations and direct links → Response displayed in the chat window → Conversation history is stored in ChatBotConversations + ChatBotMessages.

22.2 Architecture

v1: External API Integration

- ✓ AI provider: External LLM API (OpenAI GPT / Anthropic Claude). Provider is abstracted behind an IChatbotProvider interface for easy swapping.
- ✓ System prompt includes: YallaJo's domain knowledge, available tour categories, pricing ranges, locations in Jordan, and instructions to only recommend tours that exist in the database.
- ✓ User's query is enriched with context before sending to the AI: user's preferred language, location (if available), past bookings (if logged in), current season, and active promotions.
- ✓ AI response is post-processed: any tour names mentioned are matched against the database and converted to clickable links. Unmatched tour names are stripped to prevent hallucinated recommendations.
- ✓ Response language: the AI responds in the same language the user writes in. Supports English and Arabic for v1.

Future v2: Custom NLP + Database Search

- ✓ Intent detection layer: classifies user queries into intents (SearchTours, AskAboutPlace, BookingHelp, GeneralInfo, Complaint, etc.).
- ✓ Entity extraction: pulls out filters (category, location, date, budget, group size, accessibility needs) from natural language.
- ✓ Database query: translated into structured queries against the tours/places/businesses tables with the extracted filters.
- ✓ Response generation: template-based for simple queries, LLM for complex/conversational responses.
- ✓ Fallback: if intent detection confidence is below 70%, fall back to the external API.

22.3 Business Rules

Chatbot Usage Rules

- ✓ Available to everyone: guests and logged-in users.
- ✓ Rate limits: Guests: max 5 messages per session (session = until browser tab is closed). Logged-in users: max 50 messages per day (resets at midnight UTC).
- ✓ When a guest hits the 5-message limit: "Sign in to continue chatting and get personalized recommendations!"
- ✓ When a logged-in user hits the 50-message limit: "You've reached today's message limit. It resets at midnight. For urgent help, contact support."
- ✓ Conversation persistence: Logged-in users' conversations are stored in ChatBotConversations and accessible in their "Chat History" page. Guests' conversations are session-only (lost on tab close).
- ✓ Maximum message length: 500 characters per message.
- ✓ Maximum conversation length: 50 messages (25 user + 25 bot). After that, the user must start a new conversation.

Safety & Content Rules

- ✓ The AI is instructed via system prompt to ONLY discuss topics related to: tourism in Jordan, YallaJo services, booking help, travel tips, and general Jordan information.
- ✓ Off-topic queries receive: "I'm YallaJo's travel assistant! I can help you find tours, places, and activities in Jordan. What would you like to explore?"
- ✓ The AI never provides: medical advice, legal advice, political opinions, or information about competitors.
- ✓ User messages are scanned for profanity/abuse. If detected, the bot responds: "Let's keep things friendly! I'm here to help you plan an amazing trip." Repeated abuse (3+ flagged messages in a session) locks the chatbot for 24 hours.
- ✓ The AI never generates fake tour details. All tour recommendations must reference actual tours from the database (enforced by post-processing).
- ✓ PII handling: the chatbot does not ask for or store credit card numbers, passwords, or government IDs. If a user shares such data, the message is flagged and the sensitive content is redacted from storage.

Chatbot Context Enrichment (Logged-in Users)

- ✓ The AI receives (read-only) context about the user: first name, preferred language, past booking categories, favorite places, upcoming bookings, and loyalty points balance.
- ✓ This enables personalized responses like: "Since you loved your Petra tour last month, you might enjoy the Wadi Rum overnight experience!"
- ✓ User can opt out of personalized chatbot responses in their privacy settings. When opted out, the AI receives no personal context.

22.4 Handoff to Human Support

Support Escalation Rules

- ✓ If the AI cannot resolve a query (e.g., billing dispute, complex complaint), it offers: "I'd like to connect you with our support team. Would you like to open a support ticket?"
- ✓ If the user agrees, the system creates a SupportTicket pre-filled with the conversation transcript and the user's details.
- ✓ Keywords that trigger the escalation suggestion: "speak to human", "real person", "manager", "complaint", "refund", "dispute", "not working", "bug".
- ✓ The chatbot clearly identifies itself as AI: first message always includes "I'm YallaJo's AI travel assistant."

22.5 Edge Cases

Edge Cases

- ⚠ AI API is down or times out (>10 seconds) → Fallback response: "I'm having trouble connecting right now. Please try again in a moment, or contact our support team directly." System retries once. If still failing, the chat icon shows a subtle "offline" indicator.
- ⚠ User writes in a language other than English or Arabic → AI attempts to respond but may produce lower quality. A message is appended: "For the best experience, try asking in English or Arabic."
- ⚠ AI recommends a tour that was just deleted/deactivated between query and response → Post-processing catches this: dead links are removed from the response. If all recommended tours are dead, the bot says: "I couldn't find matching tours right now. Try browsing our categories."
- ⚠ User asks the chatbot to make a booking → The chatbot cannot perform transactional actions in v1. It provides a direct link: "Great choice! You can book this tour here: [link]. I can't process bookings directly yet."
- ⚠ Guest user's session expires mid-conversation → Conversation is lost (guest data is session-only). If they open a new tab, they start fresh. A subtle note at the start: "Sign in to save your conversation history."
- ⚠ User tries to extract the system prompt or jailbreak the AI → System prompt injection defenses: user messages are sanitized. The AI is instructed to ignore meta-instructions. If detected, the response is: "I can only help with travel-related questions!"

23. Smart Accessibility Features (UI)

23.1 Workflow

User opens "Accessibility" in settings or profile → Toggles desired adaptations → System saves preferences to UserPreferences → Frontend detects active preferences → CSS custom properties and ARIA attributes are dynamically updated across the entire application → Preferences persist across sessions (stored server-side for logged-in users, localStorage for guests).

23.2 Available Adaptations

Feature	What It Does	Implementation
High Contrast Mode	Increases contrast ratios to WCAG AAA (7:1 minimum). Dark backgrounds, bright text, bold borders.	CSS custom properties: --bg, --text, --accent switch to high-contrast palette. Applied globally via a .high-contrast class on <body>.
Font Size Scaling	3 levels: Normal (16px base), Large (20px), Extra Large (24px).	CSS custom property --font-base scaled. All rem-based sizes adjust proportionally. Line height increases to maintain readability.
Screen Reader ARIA Enrichment	Dynamically adds aria-labels, aria-live regions, role attributes, and skip-navigation links.	JavaScript enrichment layer adds: landmark roles to all sections, aria-labels to buttons/icons, aria-live="polite" to notification areas, sr-only text for icon-only elements.
Reduced Motion Mode	Disables all CSS animations, transitions, parallax scrolling, and auto-playing carousels.	Sets prefers-reduced-motion: reduce via CSS. JS stops all auto-advancing carousels and replaces animated transitions with instant state changes.
Color Blindness Filters	3 sub-modes: Protanopia, Deuteranopia, Tritanopia. Adjusts color palette to avoid problematic color combinations.	SVG filters applied via CSS filter property on the root element. Additionally, all color-coded UI elements (map pins, status badges) get pattern/shape differentiation alongside color.

23.3 Business Rules

Accessibility Preference Rules

- ✓ Users can enable multiple adaptations simultaneously (e.g., High Contrast + Large Font + Reduced Motion).
- ✓ For logged-in users: preferences are stored in UserPreferences table (server-side) and loaded on every page.
- ✓ For guests: preferences are stored in localStorage and a cookie. If the guest signs up, localStorage preferences are migrated to their server-side UserPreferences.
- ✓ An accessibility quick-toggle is available in the page footer on every page (no need to go to settings).
- ✓ The accessibility toggle widget itself must be fully accessible: keyboard navigable, screen-reader friendly, high contrast by default.

WCAG Compliance Rules

- ✓ Baseline: entire application targets WCAG 2.1 Level AA compliance even WITHOUT accessibility mode enabled.
- ✓ With accessibility mode: the application targets WCAG 2.1 Level AAA where feasible (7:1 contrast, enhanced focus indicators, no time limits).
- ✓ All interactive elements must be keyboard-accessible (Tab, Enter, Escape, Arrow keys).
- ✓ All images must have alt text. Decorative images use alt="" (empty alt).
- ✓ All forms must have associated labels. Error messages must be programmatically associated with the invalid field.
- ✓ Focus order must be logical and follow visual layout.
- ✓ Skip navigation link ("Skip to main content") must be the first focusable element on every page.
- ✓ Color is never the ONLY indicator of state. All status indicators use icon + color + text (e.g., booking status is not just a colored dot but also has a text label and icon).

Color Blindness Specific Rules

- ✓ Map pins: in addition to color coding (blue/green/orange), each entity type gets a distinct SHAPE (circle for tours, square for places, diamond for businesses).
- ✓ Status badges: in addition to color (green=confirmed, red=cancelled), each status gets a distinct ICON (checkmark, X, clock, etc.).
- ✓ Charts and graphs: use patterns (stripes, dots, crosshatch) in addition to colors.
- ✓ The color blindness filter selection shows a preview of how the current page looks under the selected filter before applying.

23.4 Edge Cases

Edge Cases

- ⚠ User enables Extra Large font + High Contrast on a mobile device → Some layouts may overflow. CSS uses clamp() and responsive breakpoints to prevent horizontal scrolling. Tables switch to stacked card layout on small screens with large fonts.
- ⚠ User enables Reduced Motion but a critical UI element relies on animation (e.g., loading spinner) → Loading spinners are replaced with a static "Loading..." text indicator. Progress bars remain but update instantly without transition animation.
- ⚠ Screen reader user navigates the interactive map → Map component provides an alternative: a text-based list of all locations with descriptions, sorted by distance. "View as list" toggle is always available above the map.
- ⚠ Guest user sets accessibility preferences, clears cookies, then returns → Preferences are lost. On the first visit after cookie loss, the application detects system-level preferences (prefers-reduced-motion, prefers-contrast) and auto-applies them.
- ⚠ Third-party embedded content (e.g., Mapbox, payment gateway iframe) doesn't respect accessibility settings → System applies CSS filters at the container level where possible. For iframes (payment), a note is shown: "The payment screen may not reflect your accessibility settings. Contact support if you need help completing your payment."
- ⚠ User has both browser-level dark mode and YallaJo High Contrast enabled → YallaJo's High Contrast mode takes precedence over browser dark mode to avoid double-inversion artifacts.

24. Discount & Promotions Management

24.1 Discount Creation Workflow

Provider opens "Discounts" in dashboard → Creates new discount: selects type, sets value, assigns scope, sets date range, chooses visibility → Submits for approval (Status: "Pending") → Admin reviews → Approve / Reject (with reason) / Revoke (even after approval) → If approved: discount goes live based on its ValidFrom date.

24.2 Discount Types

Type	How It Works	Example	Validation
Percentage	Reduces price by X%	20% off all tours	Value: 1–100%
Fixed Amount	Reduces price by X JOD	10 JOD off	Value must be < tour price
Early Bird	% discount when booked X+ days before tour date	15% off if booked 14+ days early	DaysBeforeTour required, min 3 days
Last Minute	% discount when booked within X days of tour date	25% off if booked within 3 days	DaysBeforeTour required, max 7 days
Group Size	Discount activates when ParticipantCount >= threshold	10% off for groups of 5+	MinParticipants field required

24.3 Discount Scope & Visibility

Scope (TargetScope)	Targets	Required Fields
AllMyTours	All tours owned by the provider	ProviderId only
SpecificTour	One specific tour	TourId required
SpecificService	One specific business service item	ServiceItemId required
Category	All tours in a category	CategoryId required

Visibility	Who Sees It	How It Works
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Public	All users	Auto-applied at checkout. Shown on listing cards with badge.
Promo Code	Users who enter the code	User types code at checkout. Not visible on listings.
Subscribers Only	Users with active YallaJo+ subscription	Visible only to subscribers. Hidden from free users. Revealed on upgrade.

24.4 Business Rules

Discount Creation Rules

- ✓ Provider creates discount → Status: Pending. Admin must approve before it goes live.
- ✓ Admin can approve, reject (with reason), or revoke a previously approved discount at any time.
- ✓ When admin revokes an active discount: it immediately stops being applied to new checkouts. Existing bookings that already used the discount are not affected.
- ✓ Discount Code (for Promo Code type): 4–20 alphanumeric characters, case-insensitive. Must be unique across all active discounts platform-wide.
- ✓ DiscountValue: must be > 0. For percentage type, max 95% (cannot create a 100% discount).
- ✓ MinOrderAmount (optional): minimum booking total before discount applies (e.g., "10 JOD off orders over 50 JOD").
- ✓ MaxDiscountAmount (optional, for percentage type): caps the discount in absolute terms (e.g., "20% off, up to 30 JOD max").
- ✓ MaxUsageCount (optional): total number of times the discount can be used across all users. When reached, discount auto-deactivates.
- ✓ MaxUsesPerUser (optional): how many times a single user can use the discount. Default: 1.
- ✓ ValidFrom / ValidTo: date range when the discount is active. Cannot be in the past on creation. ValidTo must be after ValidFrom.
- ✓ A provider can have a maximum of 20 active discounts at any time (prevents spam/abuse).

Platform Minimum Price Rule

- ✓ After ALL discounts are applied, the final booking price must be at least 5 JOD.
- ✓ If the calculated discounted price falls below 5 JOD, the system caps the discount so the final price equals exactly 5 JOD.
- ✓ This rule is enforced at checkout, not at discount creation time (because the discount might apply to tours of varying prices).
- ✓ The 5 JOD minimum is admin-configurable in system settings.
- ✓ The minimum applies to the per-booking total, not per-participant. A booking for 5 people at 1 JOD each = 5 JOD total, which meets the minimum.

Discount Stacking Rules

- ✓ Maximum 2 discounts can stack on a single booking.
- ✓ Stacking priority: the system picks the best combination of up to 2 eligible discounts that gives the user the maximum savings.
- ✓ Stacking logic: the first discount is applied to the original price. The second discount is applied to the already-discounted price (sequential, not parallel).
- ✓ Promo Code counts as one of the 2 stackable discounts. If a user enters a promo code and there's a better public discount available, both are applied (up to the 2-discount cap).
- ✓ Loyalty point redemption does NOT count as a discount for stacking purposes. Points are applied AFTER all discounts, on the final discounted price.
- ✓ The referral discount (first-booking discount for referred users) also does NOT count as a discount for stacking purposes. It's applied after discounts but before loyalty points.
- ✓ Application order: Original Price → Discount 1 → Discount 2 → Referral Discount → Loyalty Points → Final Price (must be ≥ 5 JOD).

24.5 Checkout Discount Application Workflow

User reaches checkout → System auto-evaluates all eligible Public, Early Bird, Last Minute, and Group Size discounts for the selected tour/service → Best 1–2 auto-discounts are pre-applied → User may optionally enter a Promo Code → System validates: is it active? not expired? under max uses? user eligible? → If valid, promo code is added (replacing the weaker auto-discount if the 2-cap is reached) → Discounted price shown with savings breakdown (e.g., "You save 15 JOD") → User confirms and pays → On payment success: DiscountUsage records created, usage counters incremented.

Checkout Validation Rules

- ✓ Discount eligibility is re-validated at payment time (not just at display time) to prevent race conditions where a discount expires or reaches max uses between page load and payment.
- ✓ If a discount becomes invalid between display and payment, the user is shown the updated price: "A discount has expired. Your updated total is X JOD. Proceed?"
- ✓ DiscountUsage records: DisputId, DiscountId, UserId, BookingId/ReservationId, DiscountAmount, AppliedAt.
- ✓ CurrentUsageCount on the Discount is incremented atomically (using database-level increment, not read-modify-write) to prevent overselling.
- ✓ If two users try to use the last remaining usage of a discount simultaneously, optimistic concurrency ensures only one succeeds. The other receives: "This discount has reached its usage limit."

24.6 Payout Impact

Discount Impact on Payouts

- ✓ Platform commission is calculated on the discounted price (the amount the user actually paid), NOT the original price.
- ✓ Example: Tour price 100 JOD. 20% discount applied = user pays 80 JOD. Commission (10%) = 8 JOD. Provider receives 72 JOD.
- ✓ The provider bears the cost of their own discounts. The platform does not subsidize provider-created discounts.
- ✓ Admin-created discounts (platform-wide promotions): the platform absorbs the discount cost. Provider receives payout based on the original undiscounted price minus commission.
- ✓ Discount type is tracked on the DiscountUsage record to correctly attribute who bears the cost during payout calculation.

24.7 Wishlist & Notification Integration

Wishlist Notification Rules

- ✓ When a provider activates a discount on a tour or service and admin approves it, the system queries Favorites for all users who wishlisted that entity.
- ✓ Each matching user receives a notification: "A tour on your wishlist just went on sale! [Tour Name] is now [X]% off."
- ✓ Notification channels: per user's NotificationPreferences (Email + In-App).
- ✓ Notification is sent once per discount approval. Subsequent edits to the discount (value change, date extension) do NOT trigger a new notification.
- ✓ If the discount visibility is Subscribers-Only, only users with active subscriptions who wishlisted the item are notified.
- ✓ Rate limit: max 3 wishlist-discount notifications per user per day.

Frontend Display Rules

- ✓ Active discounts show a badge on the listing card: "20% OFF", "Early Bird", "Last Minute", "Promo Code Available", etc.
- ✓ Subscribers-Only discounts show the badge ONLY to subscribers. Non-subscribers see the original price.
- ✓ When a non-subscriber views a listing with a Subscribers-Only discount, they see a teaser: "Subscribe to YallaJo+ to unlock exclusive discounts on this tour!"
- ✓ Tour detail page shows: original price (strikethrough), discounted price, savings amount, and which discount is applied.
- ✓ If multiple discounts apply, the breakdown shows each: "Early Bird: -15 JOD, Promo SUMMER25: -10 JOD. Total savings: 25 JOD."
- ✓ SalePrice field on Tour/ServiceItem entities is updated when a discount is active (for fast frontend rendering without computing discounts on every page load). Recalculated by a background job when discounts activate/deactivate/expire.

24.8 Discount State Machine

Current State	Action	Next State
(none)	Provider creates discount	Pending
Pending	Admin approves	Approved (Active when ValidFrom <= now <= ValidTo)
Pending	Admin rejects	Rejected
Rejected	Provider edits and resubmits	Pending
Approved	ValidFrom date arrives	Active
Active	ValidTo date passes	Expired
Active	MaxUsageCount reached	Exhausted
Active	Admin revokes	Revoked
Active	Provider deactivates	Inactive (provider can reactivate → Pending for re-approval)
Revoked	(terminal state)	—
Expired	(terminal state)	—

24.9 Edge Cases

Edge Cases

- ⚠ Provider creates a 95% discount on a 10 JOD tour → Discounted price = 0.50 JOD, which is below the 5 JOD platform minimum. System caps the discount at checkout so the user pays exactly 5 JOD. Savings displayed: "You save 5 JOD" (not 9.50 JOD).
- ⚠ Two discounts stack and push the price below 5 JOD → Second discount is reduced so the final price = 5 JOD. If Discount 1 alone brings it to 5 JOD, Discount 2 is not applied at all.
- ⚠ User enters a promo code that's better than both auto-applied discounts → System replaces the weaker auto-discount with the promo code, keeping the stronger auto-discount + promo code as the 2-stack.
- ⚠ Discount expires at midnight but user added the tour to cart at 23:58 → Discount is re-validated at payment time. If payment happens at 00:01, the discount is invalid and the user sees the updated price.

- ⚠ Provider creates a discount, admin approves, then provider edits the value (e.g., increases from 10% to 50%) → Edit resets status to Pending. The discount is inactive until admin re-approves the edited version.
- ⚠ Admin revokes a discount that's currently being used in 5 active checkouts (slot locks) → Revocation takes effect for NEW checkouts. The 5 users with active slot locks can still complete their booking at the discounted price (their price was locked when they entered checkout).
- ⚠ Provider has 20 active discounts (max) and tries to create another → Blocked: "You've reached the maximum of 20 active discounts. Deactivate or wait for one to expire before creating a new one."
- ⚠ User has a promo code for a Subscribers-Only discount but is not a subscriber → Promo code validation fails: "This promo code is only available to YallaJo+ subscribers. Upgrade to use it!"
- ⚠ Discount is set to TargetScope = Category, but the category is later deleted by admin → Discount is auto-deactivated (Inactive). Provider is notified: "Your discount [X] was deactivated because the category [Y] no longer exists."
- ⚠ User applies a discount, then cancels and rebooks the same tour → MaxUsesPerUser is checked per user per discount. If they used their 1 allowed use, they cannot apply it again even on the rebooking. The original DiscountUsage record persists even after cancellation.

24.10 Analytics & Reporting

Discount Analytics (Provider Dashboard)

- ✓ Providers see a "Discount Performance" section in their dashboard showing:
 - ✓ — Total uses per discount (CurrentUsageCount / MaxUsageCount).
 - ✓ — Revenue generated from discounted bookings.
 - ✓ — Conversion rate: % of users who saw the discount and completed a booking.
 - ✓ — Top-performing discount (by revenue).
- ✓ Admin sees a global discount report: platform-wide discount usage, total discount amount absorbed by providers vs. platform, abuse flags (e.g., discounts with suspiciously high usage from same IP ranges).